GAP PROMPT REVISION

GAP_FILTER_PROMPT_TEMPLATE + GAP_ANSWER_PROMPT_TEMPLATE

tap_tap_gap_answers.json Generation — Bugs Fixed, Rules Hardened

OAPI Phase A · prompts.py · Step 4

Metric	Value
Root cause	GAP_ANSWER_PROMPT_TEMPLATE allows non-OASIS EHR narrative codes and contains broken arithmetic rules
Files affected	tap_tap_gap_answers.json → oasis_gold_standard.json (cascades downstream)
Bugs in current filter prompt	2
Bugs in current answer prompt	6 (4 structural, 2 arithmetic)
Total confirmed wrong fields in SYN_0001	53 non-OASIS + 3 arithmetic errors + 2 N0415 flags
New placeholder added	{medication_json} — required for deterministic N0415 derivation
Lines changed (filter)	~5 lines hardened
Lines changed (answer)	~80 lines rewritten

1. Root Cause Analysis

The gold standard file (oasis_gold_standard.json) explicitly states it uses tap_tap_gap_answers.json as its authoritative source. Any error in the gap file propagates directly into the gold standard, which is the training signal for the OAPI optimizer. Three independent bug categories were identified.

#	Category	Effect	Severity
BUG-1	**53 non-OASIS EHR narrative codes present**	Pipeline can't match these keys — silently drops 53 training slots. Prompt item 5 ('Non-OASIS narrative codes') is the direct cause.	**CRITICAL — data loss**
BUG-2	**Generic GG0130 / GG0170 root codes used**	14 GG questions lack sub-item suffix (A1, C2, etc.). Gold standard can't map activity-specific answers — produces wrong or missing GG values.	**CRITICAL — wrong training signal**
BUG-3	**Generic GG0110 root code used**	Device/aid flags written as single code GG0110 instead of GG0110A–GG0110F sub-codes. Gold standard receives one value for 6 fields.	**HIGH**
BUG-4	**Generic N0415 root code used with array answer**	N0415 written as array of strings instead of N0415A–N0415I binary flags. Gold standard interprets incorrectly → all N0415 sub-fields wrong.	**CRITICAL — wrong training signal**
BUG-5	**PHQ D0160 computed ignoring PHQ-2 gate**	screen=2 (negative), but D0160=4 in gap file instead of 2. Gold standard imports the wrong total — H-5 mismatch.	**HIGH — arithmetic violation**
BUG-6	**C0400 grouping code set independently of sub-scores**	C0400A+B+C = 6 but C0400 = 5 in gap answers. Two-level BIMS arithmetic broken at source.	**HIGH — arithmetic violation**

2. GAP_FILTER_PROMPT_TEMPLATE — Changes

The filter prompt's job is to identify which OASIS codes are already answerable from documents. It had two gaps that allowed invalid codes to pass through into the answer phase.

Field / Rule	Status	What Changed & Why	OLD	NEW
PHQ exclusion	TIGHTENED	Added explicit exclusion of root D0150A1–D0150I2 and D0160. Old prompt only said 'PHQ mood codes' which the LLM sometimes interpreted loosely.	D0150A1 through D0150I2, D0160, PHQ_MOOD_INTERVIEW	D0150A1 through D0150I2, D0160 (PHQ MOOD INTERVIEW removed — not an OASIS code)
GG root codes	FIXED	Old rule said 'NEVER include GG discharge goal codes (GG0100, GG0110, GG0130, GG0170, GG0170C)' — this listed root codes but didn't prevent the LLM from outputting them as answerable. New rule explicitly bans root codes without full sub-item suffix.	NEVER include GG0100, GG0110, GG0130, GG0170, GG0170C	NEVER include root codes GG0130 or GG0170 without full sub-item suffix (letter+digit)
Non-OASIS gate	ADDED	No rule existed to prevent the filter from returning EHR narrative codes like ALLERGIES, CIRCULATORY_HISTORY, etc. Added explicit hard prohibition.	(absent)	Return ONLY real OASIS-E1 item codes — codes like ALLERGIES, VITAL_SIGNS, CIRCULATORY HISTORY... must NEVER appear

Side-by-Side: Filter Prompt

✕ CURRENT (BUGGY)	✓ CORRECTED
RULES: - Return ONLY clearly answerable codes - NEVER include BIMS codes (C0100 … C1310) - NEVER include PHQ mood codes (D0150A1 … D0150I2, D0160, PHQ_MOOD_INTERVIEW) - NEVER include GG discharge goal codes (GG0100, GG0110, GG0130, GG0170, GG0170C) ✕**No rule blocking EHR narrative codes** ✕LLM allowed to return ALLERGIES, SKIN, etc.	**RULES:** - Return ONLY clearly answerable codes - NEVER include BIMS codes (C0100 … C1310) - NEVER include PHQ mood codes (D0150A1 … D0150I2, D0160) ← PHQ_MOOD_INTERVIEW removed - NEVER include GG discharge goal codes ending in 2 (e.g. GG0130A2, GG0170C2) - NEVER include root GG0130 or GG0170 without suffix ✓**Hard prohibition on all non-OASIS keys:** 'Return ONLY real OASIS-E1 item codes — codes like ALLERGIES, CIRCULATORY_HISTORY, MENTAL_STATUS... must NEVER appear'

3. GAP_ANSWER_PROMPT_TEMPLATE — All Changes

Section / Rule	Status	What Changed & Why	OLD	NEW
Scope rule (preamble)	**ADDED**	No upfront scope constraint existed. LLM freely generated non-OASIS narrative codes. New CRITICAL SCOPE RULE block at top of prompt explicitly lists banned code patterns and states 'skip it entirely' for any such field.	(absent)	□ CRITICAL SCOPE RULE block — lists 15+ example banned codes, instruction to omit non-OASIS fields entirely from output
{medication_json}	**ADDED**	N0415 flags require reading medication_list.json. Old prompt had no medication context — LLM guessed from narrative and got Warfarin/Digoxin flags wrong. New placeholder passes the full medication JSON.	(placeholder absent)	--- MEDICATION LIST (use this to derive N0415 flags) --- {medication_json}
BIMS — C0300/C0400	**FIXED**	Old RULE 1 told LLM to compute C0500 = C0200+C0300A+C0300B+C0300C+C0400A+C0400B+C0400C directly, bypassing the two grouping codes. LLM set C0400=5 while C0400A+B+C=6. New rule enforces two-step arithmetic: first derive C0300 and C0400 from sub-codes, then C0500 = C0200+C0300+C0400.	C0500 MUST EQUAL EXACTLY: C0200+C0300A+C0300B+C0300C +C0400A+C0400B+C0400C	Step A: C0300 = C0300A+C0300B+C0300C Step B: C0400 = C0400A+C0400B+C0400C Step C: C0500 = C0200+C0300+C0400
PHQ D0160 formula	**FIXED**	Old RULE 2 described the gate correctly but the D0160 formula for negative screen was ambiguous. LLM computed D0160=4 when screen=2. New rule uses explicit conditional formula with worked example showing D0160=A2+B2 on negative screen.	D0160 = D0150A2 (if A1=1) + D0150B2 (if B1=1) ONLY — items C through I excluded	D0160 = (D0150A2 if D0150A1=1 else 0) + (D0150B2 if D0150B1=1 else 0) With worked example: A1=1,A2=2,B1=1, B2=1 → screen=2 → D0160=3
GG0130 sub-codes	**FIXED**	Old RULE 3 listed GG0170 sub-codes for discharge goals but GG0130 self-care items had no explicit sub-code expansion. LLM used generic GG0130 for Dressing Upper/Lower Body. New rule enumerates all GG0130 sub-items A1–G2 with activity names.	GG Discharge Goals (GG0170 items — MANDATORY)	RULE 3: NEVER output GG0130 or GG0170 root codes. Full enumeration: GG0130A1/A2=Eating, B1/B2=Oral Hygiene, C1/C2=Shower, D1/D2=Upper Dress, E1/E2=Lower Dress, F1/F2=Toileting, G1/G2=Oral Hygiene
GG0170 sub-codes	**TIGHTENED**	Old RULE 3 was discharge-goal-only for GG0170. New rule enumerates all 16 mobility activities with both admission (X1) and discharge goal (X2) codes, plus GG0170RR1 as admission-only special case.	Scale only: 01=Dependent … 88=Not attempted	Full enumeration: A1/A2=Roll Left/Right through P1/P2=Picking Up Object, plus GG0170RR1 (Wheel 50ft, admission only)
GG0110 sub-codes	**ADDED**	GG0110 (prior devices) had no expansion rule. LLM wrote GG0110 as a single root	(absent — GG0110 undefined)	RULE 5: GG0110A=Cane/Cr

		code. New RULE 5 enumerates GG0110A–GG0110F with 0=No/1=Yes semantics.		utch, B=Walker, C=Wheelchair, D=Prosthetics, E=None of above, F=Other
N0415 sub-codes	**FIXED**	Old prompt item 5 permitted non-OASIS narrative codes and had no N0415 rule. LLM wrote N0415 as array of strings. New RULE 6 derives each A–I flag deterministically from medication_json, with explicit medication→flag mapping table.	N0415: (no rule — generated as array of drug names)	RULE 6: derive N0415A-N0415I as 0/1 flags from medication_json is_high_risk fields. Anticoagulant→N0415B, Digoxin→N0415F, Insulin→N0415E, Opioid→N0415H, etc.
Non-OASIS item 5	**REMOVED**	Old item 5 'Non-OASIS narrative codes (ALLERGIES, VITAL_SIGNS, ACTIVITIES_PERMITTED, etc.): Use descriptive string answers' is the direct cause of 53 invalid fields. Entirely removed.	5. Non-OASIS narrative codes (ALLERGIES, VITAL_SIGNS, ACTIVITIES_PERMITTED, etc.): Use descriptive string answers consistent with archetype.	(removed entirely)
Output instruction	**TIGHTENED**	Old instruction said 'Every field code in the list MUST have an entry'. New instruction adds 'Non-OASIS codes must be omitted entirely — do not include them with null answers' to prevent LLM from padding output with invalid keys.	Every field code in the list MUST have an entry in the output.	Every OASIS-E1 field code MUST have an entry. Non-OASIS codes must be omitted entirely — do not include them with null answers.

4. Arithmetic Rules — Before vs. After

Three sets of calculated fields must follow strict arithmetic. The current prompt gets all three wrong in ways that cascade into the gold standard.

4.1 BIMS Two-Level Arithmetic

✗ OLD — Single-pass formula	✓ NEW — Two-step derivation
C0500 MUST EQUAL EXACTLY:	`Step A: C0300 = C0300A + C0300B + C0300C`
`C0200 + C0300A + C0300B + C0300C`	(range 0–6)
`+ C0400A + C0400B + C0400C`	`Step B: C0400 = C0400A + C0400B + C0400C`
	(range 0–6)
Problem: Skips deriving C0300 and C0400 first.	`Step C: C0500 = C0200 + C0300 + C0400`
LLM sets C0400 independently → C0400=5	(range 0–15, BIMS Total)
while C0400A+B+C = 6. Inconsistent.	
C0500 appears correct only because LLM	C0300 and C0400 are DERIVED — do NOT
happens to add sub-codes directly.	set them independently.

4.2 PHQ-2 Gate — D0160 Formula

✗ OLD — Ambiguous formula	✓ NEW — Explicit conditional with example
`D0160 = D0150A2 (if A1=1) + D0150B2 (if B1=1)`	`D0160 = (D0150A2 if D0150A1=1 else 0)`
ONLY — items C through I excluded	`+ (D0150B2 if D0150B1=1 else 0)`
Result for SYN_0001:	**Worked example (screen negative):**
screen=2 (negative), A2=1, B2=1	A1=1, A2=2, B1=1, B2=1 → screen=2
Correct D0160 = 2	`D0160 = A2(2) + B2(1) = 3`
gap_answers generated D0160 = 4 ✗	
(LLM summed extra frequency values)	NEVER include frequencies for items
	C–I when screen_score < 3.

5. N0415 High-Risk Medication Flags — Old vs. New

This is the highest-impact change. Previously the prompt had no N0415 rule at all, and permitted narrative codes — so the LLM wrote N0415 as an array of drug names. The corrected prompt replaces this with a deterministic lookup table driven by medication_list.json.

✗ CURRENT OUTPUT (SYN_0001)	✓ CORRECTED OUTPUT
`// In tap_tap_gap_answers.json:`	`// In tap_tap_gap_answers.json:`
`{`	`{ "N0415A": "0" },  // no antipsychotic`
`  "id": "n0415",`	`{ "N0415B": "1" },  // ✓Warfarin`
`  "field_codes": ["N0415"],`	`{ "N0415C": "0" },  // no antibiotic`
`  "answer": ["Anticoagulant - Warfarin", "Digoxin"]`	`{ "N0415D": "0" },  // no antiplatelet`
`}`	`{ "N0415E": "0" },  // no insulin here`
	`{ "N0415F": "1" },  // ✓Digoxin`
`// In oasis_gold_standard.json:`	`{ "N0415G": "1" },  // Furosemide`
`"N0415E": "0",  // ✗anticoagulant missing`	`{ "N0415H": "0" },  // no opioid`
`"N0415F": "0",  // ✗digoxin missing`	`{ "N0415I": "0" }  // not "none"`

N0415 Mapping Table (new RULE 6)

Code	Drug Class	Triggered by (read from medication_json is_high_risk=true)
N0415A	Antipsychotic	haloperidol, quetiapine, risperidone, olanzapine, aripiprazole
N0415B	Anticoagulant	warfarin, apixaban, rivaroxaban, dabigatran, enoxaparin, heparin (NOT aspirin/clopidogrel)
N0415C	Antibiotic	any systemic antibiotic (amoxicillin, ciprofloxacin, vancomycin, etc.)
N0415D	Antiplatelet	aspirin ≥325 mg, clopidogrel, ticagrelor, prasugrel
N0415E	Hypoglycemic	any insulin, metformin, glipizide, sitagliptin, empagliflozin, etc.
N0415F	Cardiovascular	digoxin, amiodarone, flecainide, sotalol, dronedarone (narrow TI agents)
N0415G	Diuretic	furosemide, torsemide, bumetanide, hydrochlorothiazide, chlorthalidone
N0415H	Opioid	oxycodone, morphine, hydromorphone, fentanyl, tramadol, codeine, buprenorphine
N0415I	None of above	Set '1' ONLY if ALL of N0415A–N0415H are '0'

6. GG Sub-Code Expansion — Before vs. After

The current gap file has 14 occurrences of generic root codes (GG0130, GG0170, GG0110) without sub-item suffixes. The gold standard cannot map these to specific activities. The corrected prompt fully enumerates all required sub-codes.

Root Code Used	Should Be	Activity (admission X1)	Discharge Goal (X2)
GG0130 (×2)	GG0130D1 / GG0130E1	Upper Body Dressing / Lower Body Dressing	GG0130D2 / GG0130E2
GG0110	GG0110A–GG0110F	Cane, Walker, Wheelchair, Prosthetics, None, Other	(each 0/1 flag, no discharge goal)
GG0170 (×10)	GG0170A1–GG0170P1 + GG0170RR1	Roll L/R, Sit↔Lie, Lying→Sitting, Sit→Stand, Chair Transfer, Toilet Transfer, Car Transfer, Walk 10/50/150ft, Uneven Surface, 1/4/12 Steps, Pickup, Wheel	GG0170A2–GG0170P2 (no RR2)
GG0170C	GG0170C1 / GG0170C2	Lying to Sitting on Side of Bed	GG0170C2

7. Implementation Notes

7.1 New Placeholder {medication_json}

The corrected GAP_ANSWER_PROMPT_TEMPLATE adds one new required placeholder. Your pipeline code must be updated to pass it.

```
# OLD call site (pseudo-code)
prompt = GAP_ANSWER_PROMPT_TEMPLATE.format(
    archetype=archetype,
    diagnosis_context=diagnosis_context,
    has_ambient_scribe=has_ambient_scribe,
    referral_text=referral_text,
    scribe_section=scribe_section,
    fields_with_metadata_json=fields_with_metadata_json,
)
```

```
# NEW call site — add medication_json
import json
med_json_str = json.dumps(medication_list_dict, indent=2)

prompt = GAP_ANSWER_PROMPT_TEMPLATE.format(
    archetype=archetype,
    diagnosis_context=diagnosis_context,
    has_ambient_scribe=has_ambient_scribe,
    referral_text=referral_text,
    scribe_section=scribe_section,
    medication_json=med_json_str,       # ← NEW
    fields_with_metadata_json=fields_with_metadata_json,
)
```

7.2 Recommended Post-Processing Validation

The arithmetic rules (BIMS, PHQ-2) are safety-critical. Even with the corrected prompt, a post-processing validator in Python is strongly recommended as a backstop. The LLM output should be checked before saving.

```
def validate_gap_answers(answers: dict) -> list[str]:
    errors = []
    # BIMS two-level arithmetic
    if all(k in answers for k in ['C0300A','C0300B','C0300C','C0300']):
        expected = int(answers['C0300A']) + int(answers['C0300B']) + int(answers['C0300C'])
        if int(answers['C0300']) != expected:
            errors.append(f'C0300={answers["C0300"]} but sub-sum={expected}')
    if all(k in answers for k in ['C0400A','C0400B','C0400C','C0400']):
        expected = int(answers['C0400A']) + int(answers['C0400B']) + int(answers['C0400C'])
        if int(answers['C0400']) != expected:
            errors.append(f'C0400={answers["C0400"]} but sub-sum={expected}')
    if all(k in answers for k in ['C0200','C0300','C0400','C0500']):
        expected = int(answers['C0200']) + int(answers['C0300']) + int(answers['C0400'])
        if int(answers['C0500']) != expected:
            errors.append(f'C0500={answers["C0500"]} but C0200+C0300+C0400={expected}')
    # PHQ-2 gate
    a1 = int(answers.get('D0150A1', 0))
    b1 = int(answers.get('D0150B1', 0))
```

```
    if a1 + b1 < 3:
        for letter in 'CDEFGHI':
            for col in ['1','2']:
                if answers.get(f'D0150{letter}{col}') is not None:
                    errors.append(f'D0150{letter}{col} must be null on negative PHQ-2
screen')
        a2 = int(answers.get('D0150A2', 0)) if a1 == 1 else 0
        b2 = int(answers.get('D0150B2', 0)) if b1 == 1 else 0
        expected_d0160 = a2 + b2
        if str(answers.get('D0160')) != str(expected_d0160):
            errors.append(f'D0160={answers["D0160"]} but should be {expected_d0160}')
    return errors
```

7.3 What This Does NOT Fix

These corrections fix the gap prompt only. The gold standard prompt (OASIS_GOLD_STANDARD_PROMPT) also has issues — specifically A1010 field misuse and PHQ_MOOD_INTERVIEW being written into the gold standard. Those will be addressed separately in the next revision.

✅ Fixed by this revision	☐ Still needs fixing (gold standard prompt)
• 53 non-OASIS EHR codes removed from gap file • GG0130 sub-codes fully enumerated (A1–G2) • GG0170 sub-codes fully enumerated (A1–P2, RR1) • GG0110 sub-codes enumerated (A–F) • N0415 now derived deterministically from med JSON • BIMS two-level arithmetic enforced in prompt • PHQ D0160 formula corrected with worked example • PHQ_MOOD_INTERVIEW removed from gap answers	• A1010 field definition error (gold standard) • PHQ_MOOD_INTERVIEW key in gold standard • N0415 sub-code propagation into gold standard • Gold standard GG discharge goal propagation • metadata.json validation_status field • Physician NPI fictional-range enforcement

Next step: apply the corrected prompts, re-generate SYN_0001 gap answers, then re-run the platform validation report to confirm all CRITICAL and HIGH issues are resolved before moving to the gold standard prompt revision.